UNIT – I

Fluid Statics, Buoyancy & Floatation

Subject : Fluid Mechanics & Hydraulic Machines (23A03402T)

> Syllabus: Dimensions & units, physical properties of fluids, pressure & its measurement (manometers), Pascal's & hydrostatic laws, buoyancy, metacentre, stability of floating and submerged bodies.

1. Introduction & Dimensions

Fluid = a substance that deforms continuously under the action of a shear stress, however small. (Liquids + gases.)

- Fluid mechanics = fluid statics (fluid at rest) + fluid kinematics (motion without forces) + fluid dynamics (motion with forces).
- Hydraulics deals with practical applications of liquids (mostly water) – pipes, pumps, turbines, channels.

Dimensions & Units

Every physical quantity is expressed in terms of fundamental (primary) dimensions: M, L, T (mass, length, time). All others are derived (secondary).

Units used: SI system – kg, m, s, N, Pa, W. 1 N = 1 kg·m/s^2 , 1 Pa = 1 N/m^2.

Quantity	Dimension	SI unit
Velocity v	$L\,T^{1}$	m/s
Acceleration a	$L\,T^{2}$	m/s^2
Discharge Q	$L^{3}\,T^{1}$	m^3/s
Force F	$M\,L\,T^{2}$	N
Pressure / stress p, τ	$M\,L^{-1}\,T^{2}$	N/m^2 (Pa)
Density ρ	$M\,L^{-3}$	kg/m^3
Specific weight w	$M\,L^{-2}\,T^{2}$	N/m^3
Dynamic viscosity μ	$M\,L^{-1}\,T^{1}$	N·s/m^2 (Pa·s)
Kinematic viscosity ν	$L^{2}\,T^{1}$	m^2/s
Surface tension σ	$M\,T^{2}$	N/m

Quantity	Dimension	SI unit
Work / Energy	$M\,L^2\,T^2$	J
Power P	$M\,L^2\,T^3$	W
Bulk modulus K	$M\,L^{-1}\,T^2$	N/m^2

2. Physical Properties of Fluids

(a) Density, Specific weight, Specific volume, Specific gravity

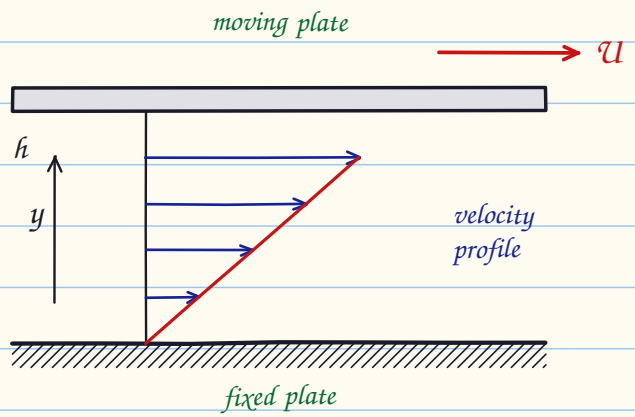
- Mass density ρ = mass / volume (kg/m^3). Water = 1000 kg/m^3 at 4°C ; Mercury = 13600 ; Air ≈ 1.2.
- Specific weight $w = \rho\, g$ = weight/volume. Water = 9810 N/m^3.
- Specific volume $v_s = 1/\rho$ (m^3/kg).
- Specific gravity (relative density) $S = \rho_{fluid} / \rho_{water\ at\ 4°C}$ – dimensionless. S for mercury = 13.6.

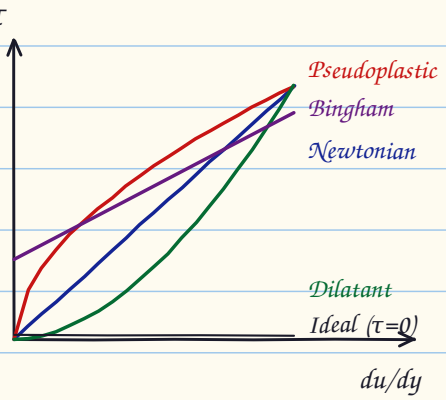
(b) Viscosity

Viscosity is the property of a fluid by virtue of which it offers resistance to relative motion (shear) between its layers. It is due to cohesion and molecular momentum exchange between layers.

Newton's law of viscosity : shear stress is proportional to the rate of shear strain (velocity gradient).

$$\tau = \mu\,(du/dy)$$





- μ = dynamic (absolute) viscosity, $N{\cdot}s/m^2$. 1 poise = 0.1 $N{\cdot}s/m^2$; 1 centipoise = 0.001 $Pa{\cdot}s$ (water ≈ 1 cP at 20°C).
- Kinematic viscosity $\nu = \mu/\rho$ (m^2/s). 1 stoke = 10^{-4} m^2/s.

- *Newtonian fluids* obey the linear law (water, air, oil). *Non-Newtonian* do not : pseudoplastic (paint, blood), dilatant (quicksand, starch), Bingham plastic (toothpaste, sewage sludge).
- *Ideal fluid* : $\mu = 0$ and incompressible. It is only a theoretical concept.

Effect of temperature

- Liquids : μ *decreases* as temperature rises (cohesion falls). Eq : $\mu = \mu_0 / (1 + \alpha t + \beta t^2)$.
- Gases : μ *increases* with temperature (molecular momentum exchange rises). Eq : $\mu = \mu_0 + \alpha t - \beta t^2$.

Significance of viscosity

Viscosity decides the frictional (shear) resistance and head loss in pipes, the lubrication film in bearings, laminar/turbulent nature of flow (Reynolds number), boundary layer and drag on bodies.

> Ex 1. Two plates are 12.5 mm apart. Upper plate moves at 2.5 m/s and needs a shear stress of 5 N/m^2. Find μ. Sol: $\tau = \mu\, U/h \Rightarrow \mu = 5 \times 0.0125 / 2.5 =$ *0.025* *$N{\cdot}s/m^2$* $(= 0.25$ poise$)$.

(c) Compressibility & Bulk modulus

Compressibility $\beta = 1/K$. Bulk modulus $K = -\,dp / (dV/V) = \rho\, dp/d\rho$.

For water $K \approx 2.07 \times 10^9$ N/m^2 (nearly incompressible); for an ideal gas, isothermal $K = p$ and adiabatic $K = k\,p$.

(d) Surface tension

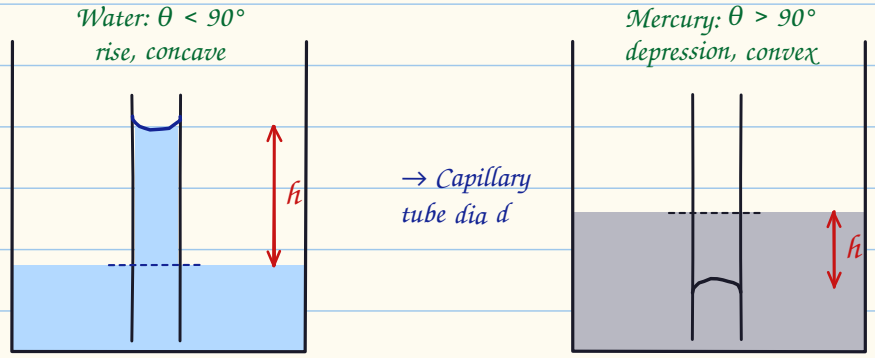
Cohesive force at the free surface/interface makes the liquid surface behave like a stretched elastic membrane. σ = force per unit length (N/m). Water–air at 20°C : $\sigma = 0.0728$ N/m.

For a surface tension force balance (pressure inside curved surface above outside):

Shape	Excess pressure p	Reason
Spherical droplet (dia d)	$p = 4\sigma/d$	one surface
Soap bubble	$p = 8\sigma/d$	two surfaces
Liquid jet (dia d)	$p = 2\sigma/d$	cylinder

(e) Capillarity

Rise or fall of a liquid in a small-bore tube relative to the surrounding level, due to cohesion and adhesion.



$$h = 4\sigma\cos\theta / (\rho g d)$$

- Adhesion > cohesion (water-glass, $\theta \approx 0°$) → rise, concave meniscus. Cohesion > adhesion (mercury-glass, $\theta \approx 130°–140°$) → fall, convex.
- $h \propto 1/d$: the thinner the tube, the higher the rise. Important in soil moisture, plant sap, blotting paper and in the error of manometer/piezometer readings (use tube dia > 12 mm).

Ex 2. Water rise in a 2 mm dia glass tube (σ = 0.0736 N/m, θ = 0°) : h = 4(0.0736)/(1000×9.81×0.002) = 0.294/19.62 = 0.015 m = 15 mm.

(f) Vapour pressure

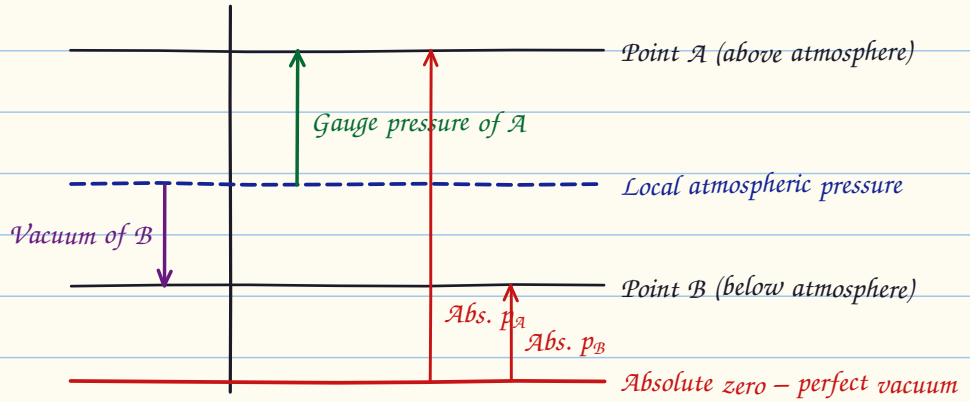
Pressure exerted by the vapour in equilibrium with its liquid at a given temperature. It rises with temperature. Water : 2.34 kPa at 20°C and 101.3 kPa at 100°C.

- A liquid boils when its absolute pressure falls to its vapour pressure.
- If local pressure in a flowing liquid (pump suction, turbine draft tube, venturi throat) falls to p_v, vapour bubbles form and collapse – cavitation : noise, vibration, pitting, loss of efficiency.

3. Pressure & its Measurement

Pressure = normal force per unit area, $p = dF/dA$ (Pa). At a point in a static fluid pressure acts equally in all directions (it is a scalar).

Atmospheric, Gauge, Vacuum and Absolute pressure



$$p_{abs} = p_{atm} + p_{gauge} \quad p_{vac} = p_{atm} - p_{abs}$$

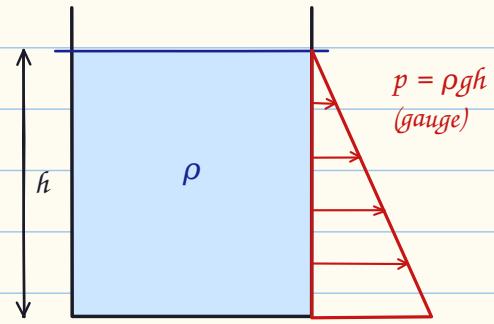
- Standard atmosphere : 1 atm = 101.325 kPa = 760 mm Hg = 10.33 m of water = 1.01325 bar.
- Absolute pressure can never be negative. Gauge pressure is negative at a point below atmosphere (= vacuum or suction pressure).
- Barometer (mercury) measures p_{atm} : $p_{atm} = \rho_{Hg}\, g\, h = 13600 \times 9.81 \times 0.76 = 101.4$ kPa.

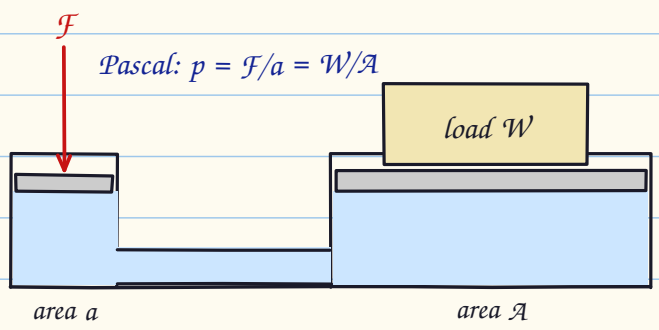
4. Pascal's Law & Hydrostatic Law

Pascal's law : intensity of pressure at a point in a fluid at rest is the same in all directions. A pressure applied to a confined fluid is transmitted undiminished everywhere.

Hydrostatic law : the rate of increase of pressure in the vertically downward direction equals the specific weight of the fluid at that point.

$$dp/dz = \rho\, g \Rightarrow p = \rho\, g\, h \text{ (gauge)}$$





- All points at the same level in a continuous static fluid are at the same pressure (horizontal planes = isobaric).
- Pressure head $h = p/(\rho g)$: the height of fluid column that produces p.
- Hydraulic press / jack / brake : $F/a = W/A \rightarrow$ mechanical advantage $= A/a$.
- Ex. Gauge pressure at 10 m depth in water $= 1000 \times 9.81 \times 10 = 98.1$ kPa ; absolute $= 98.1 + 101.3 =$ 199.4 kPa.

Total pressure & centre of pressure (plane surface)

For a plane surface of area A, centroid at depth $\bar{x}$, inclined at angle θ to the free surface :

$$F = \rho\, g\, A\, \bar{x}$$

$$h_{cp} = \bar{x} + (I_G \sin^2\theta) / (A\, \bar{x})$$

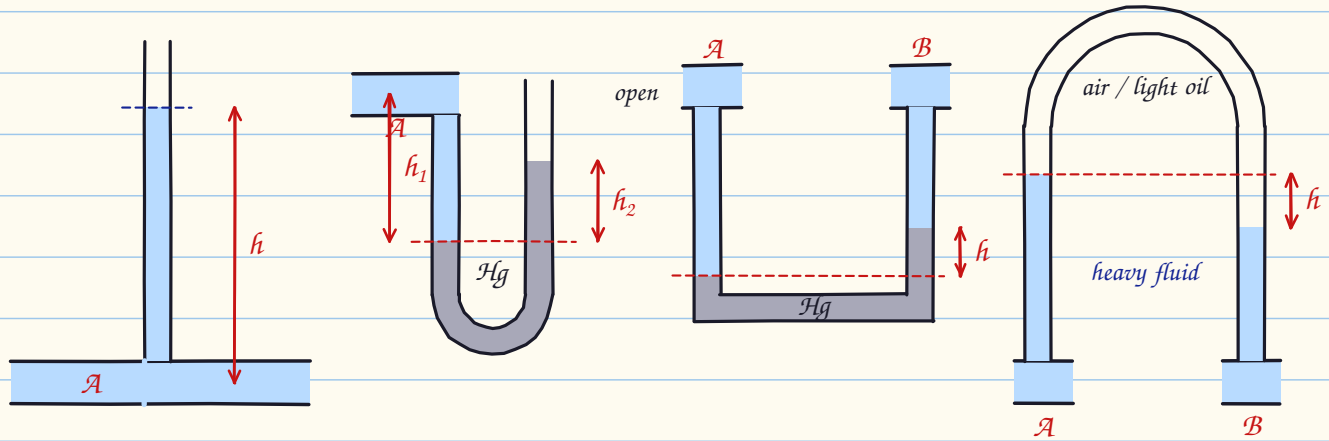
The centre of pressure always lies below the centroid (except for a horizontal surface).

> Ex 4(a). Vertical rectangular gate 2 m wide × 3 m deep, top edge at the free surface of water. $A = 6\ m^2$, $\bar{x} = 1.5$ m. $F = 1000 \times 9.81 \times 6 \times 1.5 =$ 88.3 kN. $h_{cp} = 1.5 + (2 \times 3^3/12)/(6 \times 1.5) = 1.5 + 0.5 =$ 2.0 m from the surface (= 2/3 depth).

5. Measurement of Pressure – Manometers

Manometers measure pressure by balancing it against a column of liquid (gauge liquid). Gauge liquid should be immiscible, of a known density (mercury, water, oil) and a clear meniscus.

Piezometer U-tube Differential Inverted U



(i) Piezometer

Simplest – open vertical tube tapped to the pipe. Only for liquids, only moderate gauge pressure (tube gets too tall), not for gases or negative pressures.

$$p_A = \rho g h$$

(ii) U-tube manometer

The U-tube holds a heavy gauge liquid (Hg, ρ_2) ; pipe liquid has ρ_1. Equate pressures at the common datum level (the interface in the left limb) :

$$p_A + \rho_1 g h_1 = \rho_2 g h_2 \Rightarrow p_A = \rho_2 g h_2 - \rho_1 g h_1$$

- For gas in the pipe, $\rho_1 g h_1$ is negligible $\rightarrow p_A = \rho_2 g h_2$.
- For vacuum (suction) pressure the gauge liquid rises in the pipe-side limb : $p_A = -(\rho_2 g h_2 + \rho_1 g h_1)$.

Ex 3. Pipe carries water. Centre A is 0.10 m above mercury interface (left limb); mercury in open right limb stands 0.25 m higher. $p_A = 13600(9.81)(0.25) - 1000(9.81)(0.1) = 33354 - 981 =$ 32373 $N/m^2 \approx$ 32.4 kPa (gauge).

(iii) Single-column manometer

A U-tube whose one limb is a large reservoir (area A) and the other a narrow tube (area a). Change in reservoir level is tiny, so only one limb is read.

$$p_A = \rho_2 g h_2(1 + a/A) - \rho_1 g (h_1 + a h_2/A) \; [\Delta h = a h_2/A]$$

(iv) Differential manometers (pressure DIFFERENCE between two points)

U-tube differential : A and B contain fluid ρ_1, mercury ρ_g deflects by h :

$$p_A - p_B = g h (\rho_g - \rho_1)$$

If pipes are at different levels, add $\rho_1 g (z_B - z_A)$ – i.e., measure all heads from one datum.

Inverted U-tube : used for light liquids/low differences; light fluid (air or oil, ρ_3) at top :

$$p_A - p_B = \rho_1 g h_1 - \rho_2 g h_2 - \rho_3 g h_3 \; (\text{air on top} : p_A - p_B = \rho g h)$$

Ex 3(a). Differential U-tube, pipes A and B at the same level carry water; mercury deflection $h = 0.2$ m. $p_A - p_B = g h(\rho_{Hg} - \rho_w) = 9.81(0.2)(13600 - 1000) = 24721$ Pa = 24.7 kPa (A is at the higher pressure).

(v) Micro-manometers & mechanical gauges

Inclined-tube and micromanometers amplify the reading for small pressures. Mechanical gauges (Bourdon tube, diaphragm, dead-weight tester) and pressure transducers for high pressure or remote reading.

> Method to solve any manometer problem : start from the point of known pressure, go step by step adding $\rho g h$ when moving DOWN and subtracting $\rho g h$ when moving UP, finish at the unknown point.

6. Buoyancy and Floatation

When a body is immersed (wholly or partly) in a fluid, it experiences an upward force – *buoyant force* F_B – equal to the weight of the fluid displaced (Archimedes' principle).

$$F_B = \rho_{fluid}\, g\, V_{displaced}$$

- Buoyant force acts vertically upward through the *centre of buoyancy* B = centroid of the displaced volume.
- Body weight W acts downward through the *centre of gravity* G.
- Conditions : $W > F_B$ sinks ; $W = F_B$ floats in equilibrium (partially or fully submerged) ; $W < F_B$ rises to the surface.
- For a floating body, fraction submerged = ρ_{body}/ρ_{fluid}. Iceberg (917 kg/m^3) in sea water (1025) : ≈ 89.5 % below water.
- Applications : ships, submarines, balloons/airships, hydrometers (density measurement), life jackets, buoys, pontoons.

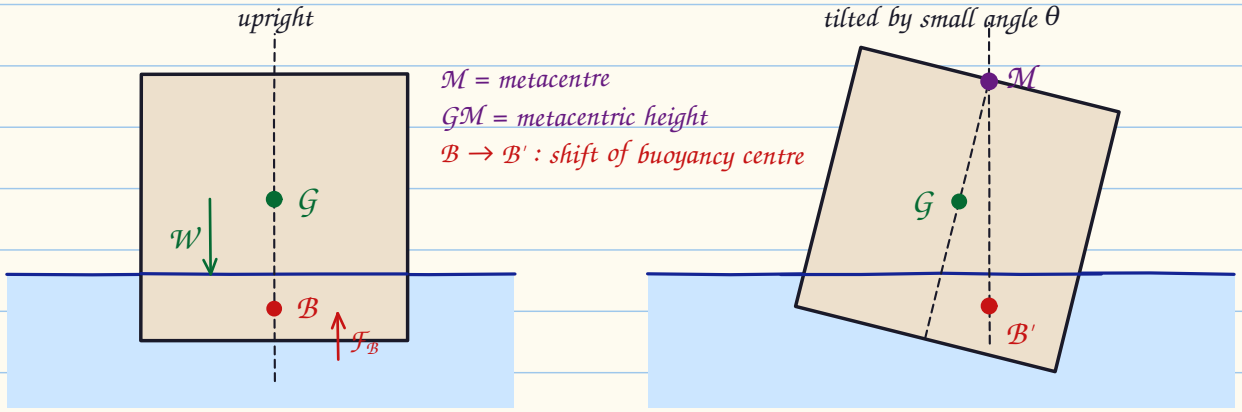
Equilibrium of submerged bodies

Position	Condition	Equilibrium
B above G	restoring couple on tilt	Stable
B below G	overturning couple	Unstable
B coincides with G	no couple (homogeneous body)	Neutral

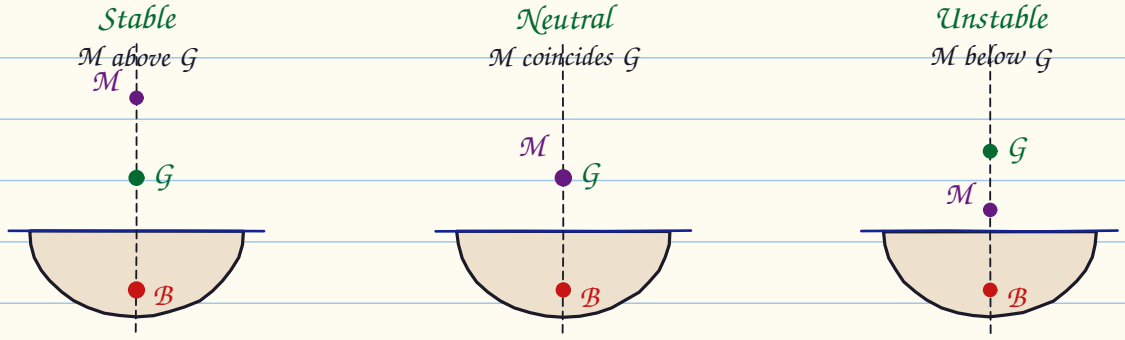
A completely submerged body keeps the same B and G whatever the tilt – stability depends only on their relative position (e.g. submarine, balloon).

7. Metacentre & Stability of Floating Bodies

In a floating body the shape of displaced volume changes when it tilts, so B shifts to B′. The vertical line through B′ cuts the original axis (BG line) at M, the metacentre.



- Metacentre M : the point about which the body starts oscillating when tilted by a small angle; point of intersection of the line of action of F_B (after tilt) with the original vertical axis.
- Metacentric height GM = distance between G and M.



Stable : M above G (GM positive) Neutral : M = G Unstable : M below G

Derivation of BM (metacentric radius)

Tilt the body by small angle θ. A wedge on one side goes into the water, an equal wedge emerges on the other side, so the buoyancy force shifts through BB′.

- Moment of the wedge forces about the axis = $F_B \times BB'$ ... a strip at distance x of width dx has volume $x\,\theta\,dA$; the moment of the wedge pair = $\rho\, g\, \theta \int x^2\, dA = \rho\, g\, \theta\, I$.
- Also moment = $F_B \times BB' = \rho\, g\, V \times BM\,\theta$ (since $BB' = BM \tan\theta \approx BM\,\theta$).

$$BM = I / V \quad GM = BM - BG = I/V - BG$$

- I = second moment of the waterline plane area about the longitudinal axis through its centroid (the axis of tilt) ; V = volume of liquid displaced.
- For rectangular waterplane ($L \times b$) : $I = L\,b^3/12$; for circular (dia d) : $I = \pi d^4/64$.
- If G is below B (BG measured downward) use $GM = BM + BG$.

Ex 4. Rectangular pontoon 10 m × 7 m × 3 m deep, draught 1.5 m (homogeneous, so G at 1.5 m above base). $I = 10\times7^3/12 = 285.8\ m^4$; $V = 10\times7\times1.5 = 105\ m^3$; $BM = 2.72$ m ; $B = 0.75$ m above base ⇒ $BG = 0.75$ m ; $GM = 2.72 - 0.75 = 1.97$ m (+ve) → STABLE.

Ex 5. Wooden cylinder, $d = 0.6$ m, height 1.0 m, $S = 0.6$, floats with axis vertical in water. Draught = $0.6 \times 1.0 = 0.6$ m ⇒ $B = 0.3$ m from base ; $G = 0.5$ m ⇒ $BG = 0.2$ m. $I = \pi(0.6)^4/64 = 0.00636\ m^4$; $V = (\pi/4)(0.6)^2(0.6) = 0.1696\ m^3$; $BM = 0.0375$ m. $GM = 0.0375 - 0.2 = -0.1625$ m → UNSTABLE (will not float upright; it floats with the axis horizontal).

Experimental determination of GM

A known weight w is moved across the deck through a distance x, causing a heel θ. Taking moments of W about G :

$$GM = w\,x / (W \tan\theta)$$

Rolling period

For small oscillations the time period of roll is

$$T = 2\pi \sqrt{(k^2 / (g \cdot GM))}\ (k = \text{radius of gyration})$$

Larger GM → shorter period : the ship is stiff (sharp, uncomfortable return). Small GM → slow roll, tender ship, closer to capsizing.

Applications & stability analysis

- Ships are designed with GM about 0.3 – 1.5 m ; low G by placing ballast/heavy cargo in the bottom, wide beam to increase I.
- Righting moment = $W \times GM \sin\theta$ (valid only for small $\theta \approx < 15°$). At large angles the GZ (righting arm) curve is used.
- Submarines : on the surface stability as a floating body (M above G); submerged – needs B above G, so heavy equipment is kept low.
- Hydrometers, buoys and floating docks are all designed using the GM criterion.

8. Quick Formula Sheet – Unit I

Quantity	Formula
Specific weight, S.G.	$w = \rho g$; $S = \rho/\rho_w$
Newton's law	$\tau = \mu\ du/dy$; $\nu = \mu/\rho$
Excess pressure	droplet $4\sigma/d$, bubble $8\sigma/d$, jet $2\sigma/d$
Capillary rise	$h = 4\sigma\ \cos\theta/(\rho\ g\ d)$
Pressure	$p = \rho\ g\ h$; $p_{abs} = p_{atm} + p_{gauge}$
Manometer	$p_A = \rho_2 g h_2 - \rho_1 g h_1$
Differential	$p_A - p_B = g\ h\ (\rho_g - \rho_1)$
Buoyancy	$F_B = \rho\ g\ V_{disp}$
Metacentre	$BM = I/V$; $GM = BM - BG$; $T = 2\pi\sqrt{(K^2/g\ GM)}$

9. Short Answers – Quick Revision

- *Is mercury a better manometer liquid than water?* Yes – high density gives a short column, low vapour pressure, clear meniscus, does not wet glass.
- *Why does a ship stay upright?* A tilt shifts B outward so F_B and W form a restoring couple whenever M lies above G.
- *Why use a wide hull and low ballast?* Wide hull ↑ I (and BM) ; low ballast lowers G ; both ↑ GM.
- *Difference between gauge and absolute pressure?* Gauge is measured relative to local atmosphere, absolute relative to perfect vacuum.
- *Why is Pascal's law useful?* Basis of hydraulic jacks, presses, brakes, lifts, excavators (force multiplication by area ratio).
- *Why is centre of pressure below centroid?* Pressure increases with depth, so the resultant acts closer to the deeper part.
- *What is a meniscus error?* Capillary rise/depression in a narrow gauge tube – avoided by using tube dia ≥ 12 mm or applying correction.
- *Vapour pressure and cavitation :* if $p \le p_v$ the liquid flashes to vapour; bubbles collapse on reaching high-pressure zones.

10. Important Questions

- Define viscosity. State Newton's law and explain the effect of temperature on viscosity of liquids and gases.
- Derive the expression for capillary rise/fall in a tube. Why does mercury show depression?
- Distinguish between absolute, gauge and vacuum pressure. Derive the hydrostatic law.
- Describe the differential U-tube and inverted U-tube manometers with neat sketches; derive the expressions.
- Define buoyancy, centre of buoyancy and metacentre. Derive $BM = I/V$.
- Explain the experimental determination of metacentric height. State the conditions of stability of floating & submerged bodies.
- Numerical : manometer problems, GM of rectangular/cylindrical vessels, force on submerged plane surfaces.